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16 September 1982
SUPERSEDING
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8 May 1964

MILITARY SPECIFICATION

STEEL, STRIP, HIGH-CARBON, SPHEROIDIZED ANNEALED (FOR AMMUNITION CARTRIDGE CLIPS)

This specification is approved for use by the US Army Material and Mechanics Research Center, ATTN: DRXMR-SSS, Watertown, MA 02172, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of cold finished carbon steel strip in the spheroidized annealed condition.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Std. No. 66 - Steel: Chemical Composition and Hardenability.
Fed. Test Method Std. No. 151 - Metals; Test Methods.
Fed. Std. No. 183 - Continuous Identification Marking of Iron and Steel Products.

MILITARY

MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SSS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

/FSC 9515/

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(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS:

ASTM A 370 - Mechanical Testing of Steel Products

ASTM E 112 - Methods for Estimating the Average Grain Size of Metals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Inudstry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of the steel shall meet the requirements of table I. A cast or heat analysis of each heat of steel shall be furnished by the contractor.

TABLE I. Chemical composition, cast or heat analysis

Element	Percent
Carbon	0.43-0.50
Manganese	0.45-0.75
Phosphorous	0.030 max.
Sulphur	0.035 max.
Silicon	0.15-0.35

3.1.1 Product analysis tolerances. Product analysis tolerances shall be in accordance with Fed. Std. No. 66.

3.2 Mechanical properties.

3.2.1 Tensile. The tensile strength of the annealed strip may be specified (see 6.2) in which case the specimens representing the steel shall meet the requirements as specified in the invitation for bids, request for proposal, on the drawing, or in the contract or order (see 6.2).

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3.2.2 Hardness. The hardness of the annealed strip may be specified (see 6.2), in which case the steel shall meet the requirements specified in the invitation for bids, request for proposal, on the drawing, or in the contract or order (see 6.2).

3.2.3 Bending. Unless otherwise specified (see 6.2), the specimens representing the strip shall withstand cold bending without cracking on the outside of the bent portion as indicated in table II.

TABLE II. Cold bending requirements

Degree of bend	Ratio inside radius to thickness	Relation of bend to rolling direction
180°	Flat back	longitudinal
180°	t/2[a]	transverse

[a] Up to 0.0062 inch (0.16 mm) thickness maximum.

3.3 Austenite grain size. The steel shall have an austenite grain size of 5 or finer. This requirement shall be met if 70 percent or more of the grains are size 5 or finer.

3.4 Condition. The strip shall be furnished in the spheroidized annealed condition and shall be free of lamellar pearlite. When specified (see 6.2), the microstructure shall conform to standards furnished by the procuring activity.

3.5 Decarburization. The strip shall not show a depth of total and partial decarburization in excess of 0.001 inch (0.025 mm), or 1.5% of the thickness of the strip, whichever is greater. The standards for depth of decarburization shall be furnished or approved by the procuring activity (see 6.2).

3.6 Hardenability. When heat treated in accordance with the appplicable cartridge clip drawing, the strip shall be capable of attaining the hardness specified in the applicable drawing.

3.7 Fabricating requirements. Strip shall be capable of withstanding the fabricating test without failure (see 4.5.9).

3.8 Dimensions and tolerances. Strip shall meet the dimensions and tolerances shown in the applicable cartridge drawing, or in the contract or order (see 6.2) and as specified below.

3.8.1 Width. Unless otherwise specified (see 6.2), the nominal width of the strip shall be 3-3/4 inches (95.3 mm).

3.8.2 Crown. Crown shall be limited so that the thickness across the width of the strip shall be within the specified tolerances.

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3.8.3 Camber. The strip shall not vary in edgewise straightness by more than 1/4 inch (6.35 mm) in an eight foot (2.4 m) length.

3.9 Finish. Unless otherwise specified (see 6.2), strip shall have a No. 2 finish, regular bright finish.

3.10 Edge. Unless otherwise specified (see 6.2), the strip shall have a No. 3 edge, slit edge (approximately square edge produced by slitting, not filed).

3.11 Identification marking. Each coil shall be marked in accordance with Fed. Std. No. 183 or as agreed upon by the contractor and procuring agency.

3.12 Workmanship. Strip shall be clean, commercially free from rust, scabs, seams, pits, blisters, laminations, and other imperfections detrimental to the specific use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all strip from the same heat, from the same annealing charge subject to inspection at the same time.

4.2.1 Austenite grain size. A lot for austenite grain size determination shall be one heat.

4.3 Sampling.

4.3.1 Chemical analysis. One sample shall be taken from at least 3 coils in each lot for product analysis by the procuring agency. Samples shall be taken in accordance with method 111 or 112 of Fed. Test Method Std. No. 151.

4.3.2 Mechanical properties.

4.3.2.1 Tensile. When tensile strength is specified (see 3.2.1), two samples shall be taken from each lot.

4.3.2.2 Hardness. When hardness requirements are specified (see 3.2.2), at least 10 percent of the coils in each lot shall be sampled for hardness testing.

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4.3.2.3 Bending. Unless otherwise specified in the invitation for bids, request for proposal, or in the contract or order (see 6.2), samples for bend testing shall be taken from at least 25 percent of the coils in each lot. Sampling may be reduced to 10 percent of the lot in the presence of evidence to the procuring activity that the method of manufacture and ductility of the strip are suitably uniform. Each sample shall be of sufficient size for one longitudinal and one transverse bend specimen and shall represent the full width of the strip.

4.3.3 Austenite grain size. One sample from each heat shall be taken for austenite grain size determination.

4.3.4 Condition. Samples for microexamination of the spheroidized annealed condition shall be taken from two coils in each lot.

4.3.5 Decarburization. Samples for decarburization determination shall be taken from two coils in each lot.

4.3.6 Hardenability. Samples for the hardenability test shall be taken from two coils in each lot.

4.3.7 Fabrication. At least 5 coils shall be taken from each lot for fabrication testing.

4.3.8 Visual and dimensional. The coils selected for the bending test samples shall be used for visual and dimensional sampling.

4.4 Examination.

4.4.1 Visual. Sample coils selected in accordance with 4.3.8 shall be examined for conformance with the requirements for finish (see 3.9), edge (see 3.10), identification marking (see 3.11) and workmanship (see 3.12).

4.4.2 Dimensions and tolerances. A representative number of measurements shall be taken on the sample coils to determine compliance with the requirements for dimensions and tolerances (see 3.8).

4.4.3 Packaging inspection. Prior to shipment, examination shall be made to determine compliance with the requirements of section 5.

4.5 Tests.

4.5.1 Test specimens.

4.5.1.1 Chemical composition. Test samples and specimens shall be prepared in accordance with method 111 or 112 of Fed. Test Method Std. No. 151, as applicable.

4.5.1.2 Tensile. Test specimens shall be in full section or shall be machined to the form and dimensions of the sheet type specimen of ASTM A 370.

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4.5.1.3 Hardness. Specimens shall be prepared in accordance with ASTM A 370.

4.5.1.4 Bending. Specimens shall be prepared in accordance with ASTM A 370. The specimens for longitudinal testing shall represent the full width of the strip and for transverse specimens shall be at least 2 inches (50.8 mm) in the direction parallel to the rolling and the full width transverse to the rolling direction.

4.5.1.5 Austenite grain size. The specimen shall be prepared in accordance with ASTM E 112.

4.5.1.6 Condition. Two specimens shall be prepared at least one inch (25.4 mm) in length and the full thickness of the strip. One specimen shall be prepared in the longitudinal direction and one in the transverse direction for metallographic examination. Specimens shall be prepared using precautions to insure freedom from disturbed metal.

4.5.1.7 Decarburization. The specimens prepared for determining spheroidization shall be used for determination of decarburization.

4.5.1.8 Hardenability. The samples shall be heat treated in accordance with the heat treating instructions on the applicable cartridge clip drawing and prepared for hardness testing in accordance with ASTM A 370.

4.5.2 Test procedures.

4.5.2.1 Chemical composition. Chemical analysis shall be conducted in accordance with method 111 or 112 of Fed. Test Method Std. No. 151 (see 6.3).

4.5.2.2 Tensile. Tensile tests, when specified (see 3.2.1), shall be conducted in accordance with ASTM A 370.

4.5.2.3 Hardness. Hardness tests, when specified (see 3.2.2), shall be conducted in accordance with ASTM A 370. Hardness readings shall be taken near the end of the sample coils.

4.5.2.4 Bending. Bend tests shall be conducted in accordance with ASTM A 370.

4.5.2.5 Austenite grain size. Austenite grain size shall be determined in accordance with ASTM E 112.

4.5.2.6 Condition. Microexamination of the prepared specimens shall be made to determine conformance with 3.4 and comparison made against the standard furnished by the procuring activity.

4.5.2.7 Decarburization. Depth of decarburization shall be determined microscopically and comparison made against the standards furnished by the procuring activity.

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4.5.2.8 Hardenability. Hardness testing on the heat treated samples shall be conducted in accordance with ASTM A 370. Hardness readings shall be taken at intervals of one inch (25.4 mm) or less across the width of the strip.

4.5.2.9 Fabrication. The sample coils shall be fabricated, to the extent necessary, including heat treatment, to determine the suitability of the strip for its intended use.

4.6 Rejection and retests.

4.6.1 Examination. If the sampling for visual, dimensional, or packaging fails to meet the requirements of this specification when examined in accordance with 4.4, the lot shall be rejected.

4.6.2 Tests. A lot shall be rejected for failure to comply with any of the specified test requirements when tested in accordance with 4.5.

4.6.3 Retests. Retests shall be permitted in accordance with Fed. Test Method Std. No. 151.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-STD-163 for the levels or degree of protection specified (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order (see 6.2), shipments shall be marked in accordance with the requirements of MIL-STD-163.

6. NOTES

6.1 Intended use. The cold finished carbon steel strip in the spheroidized annealed condition, covered in this specification, is intended for use in cartridge clips.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Tensile strength, if required (see 3.2.1).
c. Hardness, if specified (see 3.2.2).
d. Standards for spheroidized annealed condition (see 3.4).
e. Standards for decarburization (see 3.5).
f. Dimensions and tolerances, if not shown on drawing (see 3.8).
g. Finish, if other than No. 2 (see 3.9).
h. Edge, if other than No. 3 (see 3.10).
i. Lot size, if other than as specified in 4.2.
j. Levels of packaging and packing desired (see section 5).
k. When special marking for shipment requirements are necessary (see 5.2).

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6.3 Chemical analysis. The suggested instrumental method that can be used for chemical analysis is ASTM E 415. ASTM A 571 should be consulted for a complete list of methods.

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